

Setup Guide

Mobile Collaborative Robot software



Client: École Nationale d'Ingénieurs de Tarbes

Group name: BOB

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Project name: Mobile Collaborative Robot - EPS

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1. Prerequisites

Recommended System Specifications

CPU: 8-core

RAM: 16 GB (at least)

GPU: NVIDIA GTX 1650 ~ RTX 3060

OS: Ubuntu 24.04 (64-bit) - Ubuntu must be installed native or as a dual-boot system

Storage: 70GB

Note

A dedicated GPU is not strictly required, but Gazebo and RViz performance will be significantly reduced without one (lower FPS and slower rendering).

2. ROS2 installation

install **ROS2 Jazzy Jalisco**, Follow the official installation instructions on the ROS2 website ([ROS 2 Documentation Jazzy](#)).

Note

After installation, Open a terminal (Ctrl + Alt + T) and run:

```
echo $ROS_DISTRO
```

If the version is jazzy, proceed to Step 3.

3. Simulation tool installation

The following ROS2 components are required for the EPS simulation environment:

- ros2_control / ros2_controllers
- Navigation2 (Nav2)
- MoveIt2
- Gazebo-ROS integration
- Kortex driver (Kinova Gen3)
- RViz / rqt visualization tools

Refer to the Appendix for the full list of required APT packages.

4. ROS2 Packages

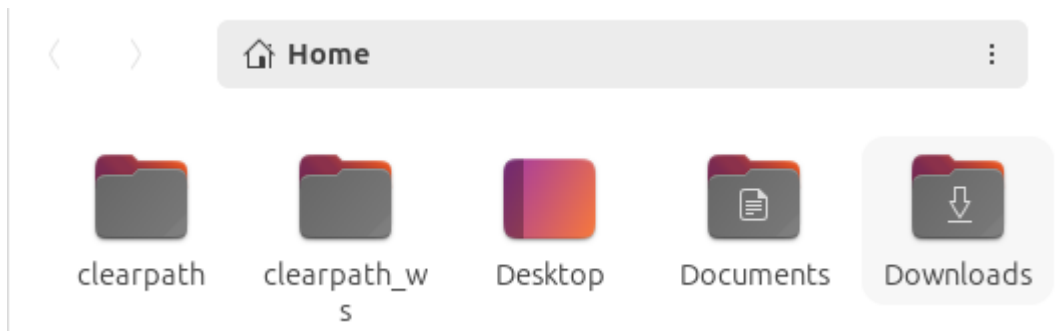
Team BOB provides a ready to use src folder.

Place the provided src into:

```
~/clearpath_ws/src
```

a. Create "Clearpath" folder in HOME

Create a folder named clearpath directly in your HOME directory NOT inside clearpath_ws.



And copy these files: robot.yaml, warehouse.pgm & warehouse.yaml



The provided src folder already contains:

- Clearpath / Ridgeback packages (simulation, description, config)
- MoveIt2 packages (motion planning)
- Kortex packages (Kinova Gen3 communication)
- EPS custom packages (bringup, mirror node, configs)
- ros2_control configuration packages



b. Build the workspace

Build the workspace:

```
cd ~/clearpath_ws
```

```
colcon build --symlink-install
```

```
source install/setup.bash
```

After building

```
echo "source ~/clearpath_ws/install/setup.bash" >> ~/.bashrc
```

the workspace (To add workspace to bashrc so it loads automatically):

```
source ~/.bashrc
```

5. Building the simulation

Before running the simulation, your workspace must be built. With the following commands, if this has already been done continue to the next step.

<code>cd ~/clearpath_ws</code>
<code>colcon build --symlink-install</code>
<code>source install/setup.bash</code>

Also make sure the clearpath folder exist in ~/HOME

6. Running the simulation

To start the full simulation (Ridgeback + Kinova Gen3), run:

```
ros2 launch eps_bringup eps_sim.launch.py
```

If everything is set correctly, Gazebo will open automatically, and the Ridgeback + Kinova Gen3 Robot will appear in the warehouse map.

7. Running Physical robot

1) Connect Ethernet cable to the robot

2) Check interface: ip a

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 14:13:33:00:cb:c9 brd ff:ff:ff:ff:ff:ff
    inet 172.28.8.249/19 brd 172.28.31.255 scope global dynamic noprefixroute wlp1s0
        valid_lft 13886sec preferred_lft 13886sec
    inet6 fe80::d76d:716c:a304:1249/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enx207bd51a0355: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 20:7b:d5:1a:03:55 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.11/24 brd 192.168.1.255 scope global noprefixroute enx207bd51a0355
        valid_lft forever preferred_lft forever
```

3) Open eps_kinova_connect.sh and modify:

IFACE="your_interface_name"

```
set -e

ROBOT_IP="192.168.1.10"
IFACE="enx207bd51a0355"
HOST_IP="192.168.1.11"
WS="$HOME/clearpath_ws"
```

4) Run:

```
chmod +x eps_kinova_connect.sh
```

```
./eps_kinova_connect.sh
```

8. Appendix A: required APT Packages

APT Packages list (Optional):

```
sudo apt install ros-jazzy-ros2-control ros-jazzy-ros2-controllers
```

```
sudo apt install ros-jazzy-navigation2 ros-jazzy-nav2-bringup
```

```
sudo apt install ros-jazzy-gz-ros-pkgs
```

```
sudo apt install ros-jazzy-moveit ros-jazzy-moveit-servo
```

```
sudo apt install ros-jazzy-rviz2 ros-jazzy-rqt*
```

9. Appendix B: References

Clearpath robotics. (n.d.). *Clearpath robotics documentation*. Retrieved December 2025, from
https://docs.clearpathrobotics.com/docs_robots/indoor_robots/ridgeback/user_manual/ridgeback/